

CODE-GRAPH — KZTEK Labeling Studio

Tài liệu bản đồ codebase. Mọi coding agent PHẢI đọc file này TRƯỚC khi mở source.
Cập nhật ngay sau mỗi PR merge có thay đổi cấu trúc/API/schema.

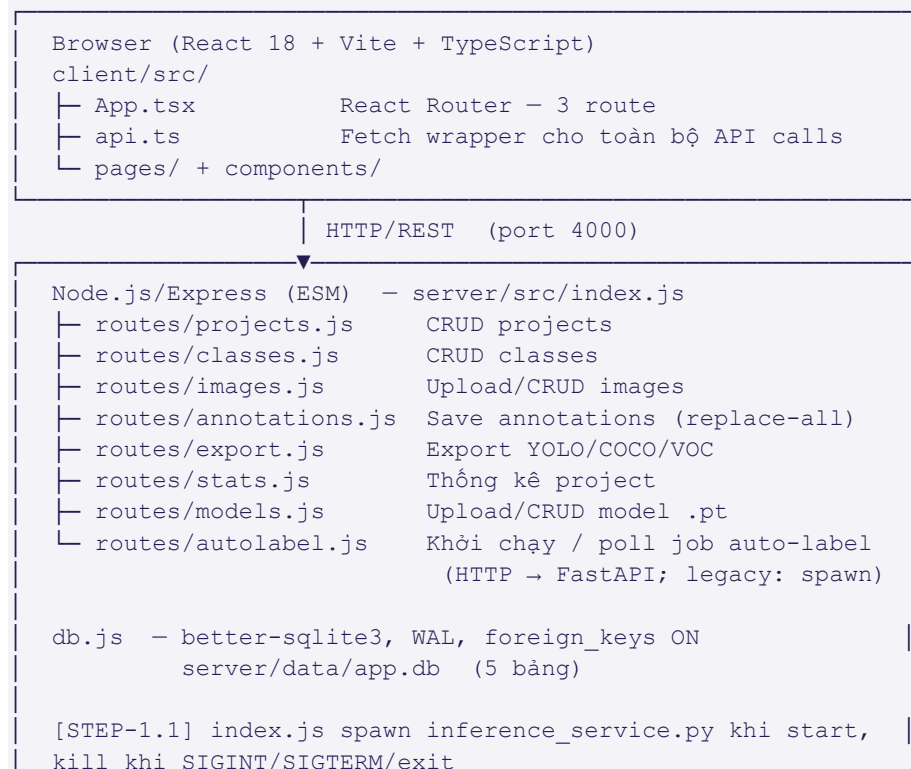
>

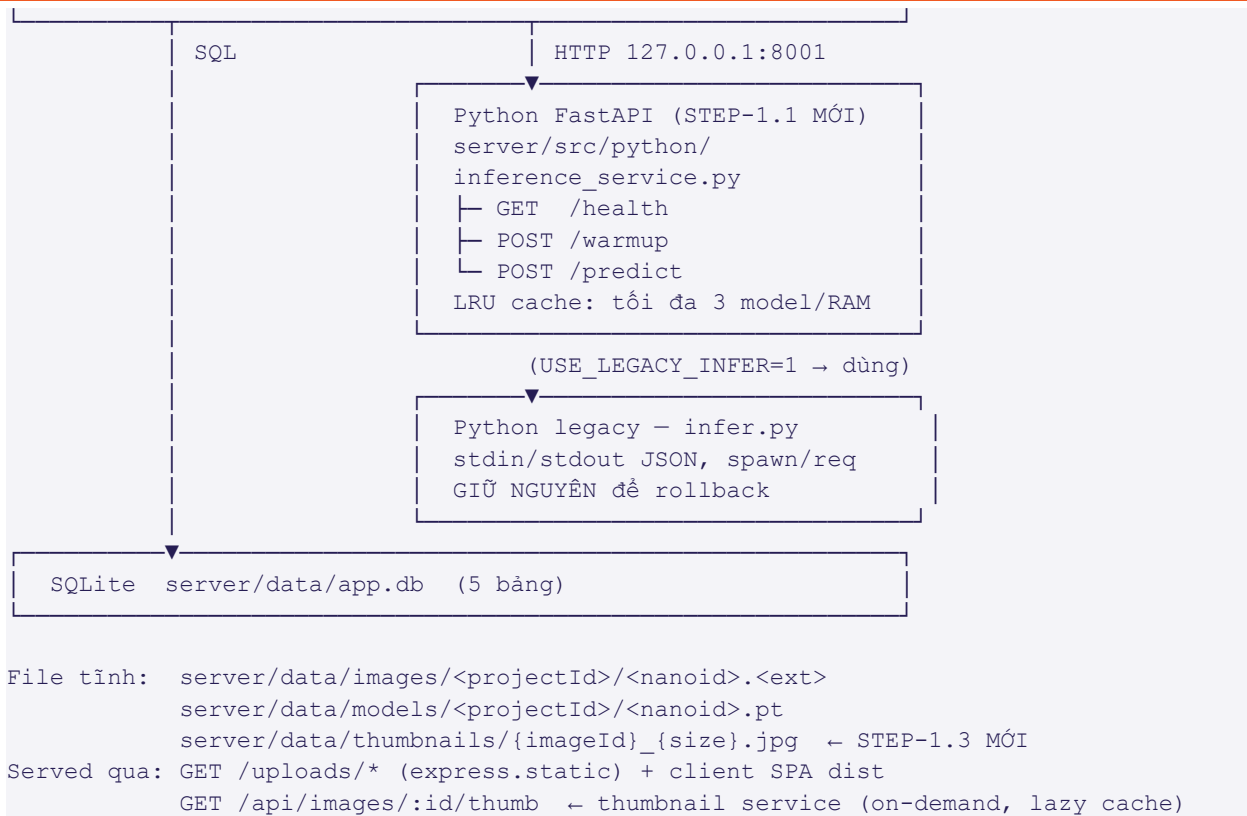
Last verified: 2026-08-05 | Cập nhật: senior-developer (STEP-4.1)

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1. Tổng quan kiến trúc





Port: 4000 (server + serve SPA dist + /uploads)

Inference port: 8001 (env `INFERENCE_PORT`, mặc định 8001)

DB file: `server/data/app.db` (tạo tự động nếu chưa có)

Python bin: env `PYTHON_BIN` hoặc `python` (mặc định)

Rollback: env `USE_LEGACY_INFER=1` → dùng `infer.py` spawn-per-request

2. Cấu trúc thư mục

Roboflow - Copy/	
├ server/	
│ └ src/	
│ │ └ index.js	← Entry point Express
│ │ └ db.js	← DB init, schema, migrations
│ │ └ routes/	
│ │ │ └ projects.js	
│ │ │ └ classes.js	
│ │ │ └ images.js	← multer, sharp, adm-zip; GET / trả thêm
└ thumbnail_url	
│ │ └ thumbnails.js	← thumbnail cache service (STEP-1.3 MỚI)
│ │ └ annotations.js	← STEP-3.2: ghi snapshot vào annotation_history sau
│ │ │ mỗi save	
│ │ │ └ history.js	← STEP-3.2 MỚI: GET /history + POST
│ │ │ └ /history/:v/revert	
│ │ │ └ activity.js	← STEP-3.3 MỚI: GET /activity (project activity log)
│ │ │ └ export.js	← archiver (yolo/coco/voc); STEP-3.3: gọi
logActivity	

```

├── stats.js
├── models.js
├── autolabel.js      ← HTTP → FastAPI (STEP-1.1); DB-backed jobs (STEP-
1.2)
├── jobs.js          ← CRUD /api/jobs (STEP-1.2 MỚI)
├── python/
│   └── inference_service.py ← FastAPI thường trực, LRU cache 3 model (STEP-
1.1 MỚI)
├── infer.py          ← stdin→stdout JSON batch inference (GIỮ LẠI,
rollback)
├── migrations/      ← SQL migration files (chỉ tham khảo; db.js inline
là nguồn sự thật)
│   └── 001_create_jobs.sql ← (STEP-1.2)
├── data/             ← RUNTIME (git-ignored)
│   ├── app.db
│   ├── images/<projectId>/
│   ├── thumbnails/   ← cache thumbnail (STEP-1.3 MỚI):
│   │   └── {imageId}_{size}.jpg
│   └── models/<projectId>/
│       └── package.json
├── client/
│   ├── src/
│   │   ├── main.tsx
│   │   ├── App.tsx      ← React Router setup
│   │   ├── api.ts        ← Tất cả fetch call
│   │   ├── types.ts      ← TS interfaces
│   │   ├── styles.css
│   │   ├── pages/
│   │   │   ├── ProjectsPage.tsx ← List/create/delete projects
│   │   │   ├── ProjectDetailPage.tsx ← Image grid, upload, filters
│   │   │   └── AnnotatorPage.tsx ← Canvas editor (bbox+quad), autosave
│   │   ├── components/
│   │   │   ├── Logo.tsx
│   │   │   ├── StatsPanel.tsx
│   │   │   ├── ExportModal.tsx
│   │   │   └── AutoLabelModal.tsx
│   │   ├── utils/
│   │   │   └── files.ts
│   └── package.json
├── code-graph/
│   ├── CODE-GRAPH.md    ← file này
│   └── CODE-GRAPH.pdf
└── docs/plans/          ← Plan files
  
```

3. Module map — Server

3.1 `server/src/index.js` — Entry point

Thuộc tính	Giá trị	Confidence		
---	---	---		
Loại	Express app entry	CONFIRMED		
Port	`process.env.PORT` \	\	4000`	CONFIRMED
Middleware	cors(), express.json(limit 10mb), express.static(/uploads)	CONFIRMED		

SPA fallback	serve <code>client/dist/index.html</code> cho mọi non-API route	CONFIRMED		
Callers/Used-by	(entry point — không ai import)	CONFIRMED		
Imports	db.js (UPLOAD_DIR), tất cả 8 route files, node:child_process (spawn)	CONFIRMED		
[STEP-1.1] Inference lifecycle	<code>startInferenceService()</code> → spawn <code>inference_service.py</code> sau <code>app.listen()</code> . <code>stopInferenceService()</code> đăng ký qua <code>process.on('exit'/'SIGINT'/'SIGTERM')</code>	CONFIRMED		
[STEP-1.2] Jobs router	import + mount <code>routes/jobs.js</code> tại <code>/api/jobs</code>	CONFIRMED		
[STEP-1.3] Thumbnails router	import + mount <code>routes/thumbnails.js</code> tại <code>/api/images</code>	CONFIRMED		
[STEP-3.2] History router	import + mount <code>routes/history.js</code> tại <code>/api/images/:imageId</code>	CONFIRMED		
[STEP-3.3] Activity router	import + mount <code>routes/activity.js</code> tại <code>/api/projects/:projectId/activity</code>	CONFIRMED		
[STEP-4.2] Prefill router	import + mount <code>routes/prefill.js</code> tại <code>/api/projects/:projectId</code>	CONFIRMED		
Env vars đọc	PORT, PYTHON_BIN, INFERENCE_PORT, USE_LEGACY_INFER	CONFIRMED		
Last verified	2026-08-05 (STEP-4.2)	-		

Route mounting:

<code>/api/projects</code>	→ <code>routes/projects.js</code>	
<code>/api/projects/:projectId/classes</code>	→ <code>routes/classes.js</code>	
<code>/api/projects/:projectId/images</code>	→ <code>routes/images.js</code>	
<code>/api/images/:imageId/annotations</code>	→ <code>routes/annotations.js</code>	
<code>/api/projects/:projectId/export</code>	→ <code>routes/export.js</code>	
<code>/api/projects/:projectId/stats</code>	→ <code>routes/stats.js</code>	
<code>/api/projects/:projectId/models</code>	→ <code>routes/models.js</code>	
<code>/api/projects/:projectId/auto-label</code>	→ <code>routes/autolabel.js</code>	
<code>/api/jobs</code>	→ <code>routes/jobs.js</code>	[STEP-1.2 MỚI]
<code>/api/images</code>	→ <code>routes/thumbnails.js</code>	[STEP-1.3 MỚI]
<code>/api/images/:imageId</code>	→ <code>routes/reviews.js</code>	[STEP-2.3 MỚI]
<code>/api/images/:imageId</code>	→ <code>routes/history.js</code>	[STEP-3.2 MỚI]
<code>/api/projects/:projectId/activity</code>	→ <code>routes/activity.js</code>	[STEP-3.3 MỚI]
<code>/api/projects/:projectId</code>	→ <code>routes/prefill.js</code>	[STEP-4.2 MỚI]

3.2 `server/src/db.js` — Database layer

Thuộc tính	Giá trị	Confidence
---	---	---
Loại	Singleton DB module (ESM export)	CONFIRMED
Exports	db, DATA_DIR, UPLOAD_DIR, MODEL_DIR, DB_PATH, HISTORY_MAX_VERSIONS, <code>pruneAnnotationHistory()</code> , <code>logActivity()</code>	CONFIRMED
Callers/Used-by	index.js (UPLOAD_DIR), projects.js,	CONFIRMED

	classes.js, images.js, annotations.js, export.js, stats.js, models.js, autolabel.js, jobs.js, history.js, activity.js	
DB engine	better-sqlite3, WAL mode, foreign_keys=ON	CONFIRMED
Schema init	CREATE TABLE IF NOT EXISTS — idempotent	CONFIRMED
Migrations inline	PRAGMA table_info + ALTER TABLE cho cột thiếu	CONFIRMED
[STEP-1.2] Startup cleanup	UPDATE jobs SET status='error' WHERE status IN ('running', 'pending') — mọi job dở dang khi restart đều bị đánh dấu error	CONFIRMED
[STEP-4.1] m007_detect_cache()	CREATE TABLE detect_cache + unique index (image_id, model_id) + 2 indexes; verify row count trước/sau	CONFIRMED
[STEP-4.2] m008_default_model_id()	ALTER TABLE projects ADD COLUMN default_model_id TEXT; verify row count projects trước/sau	CONFIRMED
Last verified	2026-08-05 (STEP-4.2)	-

Inline migrations hiện có (chạy khi server khởi động):

1. annotations.type — nếu chưa có → ADD COLUMN type TEXT DEFAULT 'bbox'
2. annotations.points — nếu chưa có → ADD COLUMN points TEXT
3. classes.hotkey — nếu chưa có → ADD COLUMN hotkey TEXT
4. [STEP-1.2] Cleanup jobs — UPDATE running/pending → error (startup idempotent)
5. [STEP-2.2] m003_add_users() — CREATE TABLE users + seed admin + ADD COLUMN images.uploaded_by
6. [STEP-2.3] m004_add_review_status() — ALTER TABLE images (4 review cols) + index
7. [STEP-3.1] m005_phase3_schema() — ADD COLUMN annotations.version + CREATE TABLE annotation_history + activity_log; backup + verify row count
8. [STEP-3.5] m006_add_image_done_columns() — ALTER TABLE images (completed_at + completed_by) + index; verify row count
9. [STEP-4.1] m007_detect_cache() — CREATE TABLE detect_cache + 3 indexes; verify row count
10. [STEP-4.2] m008_default_model_id() — ALTER TABLE projects ADD COLUMN default_model_id TEXT; verify row count

3.3 `server/src/routes/projects.js`

Thuộc tính	Giá trị	Confidence
---	---	---
Router options	Router() (không mergeParams)	CONFIRMED
Imports	db	CONFIRMED

Callers/Used-by	index.js (mounted /api/projects)	CONFIRMED
Last verified	2026-08-04	-

Method	Path	File:Line	Mô tả
---	---	---	---
GET	/	projects.js:9	List all projects (JOIN count images/labeled/classes)
POST	/	projects.js:20	Create project (auto-create 2 default classes)
GET	/:id	projects.js:33	Get project by id
PATCH	/:id	projects.js:39	Update name/description
DELETE	/:id	projects.js:48	Delete project (CASCADE images/classes/annotations)

3.4 `server/src/routes/classes.js`

Thuộc tính	Giá trị	Confidence
---	---	---
Router options	Router({ mergeParams: true })	CONFIRMED
Imports	db	CONFIRMED
Callers/Used-by	index.js (mounted /api/projects/:projectId/classes)	CONFIRMED
Last verified	2026-08-04	-

Method	Path	File:Line	Mô tả
---	---	---	---
GET	/	classes.js:13	List classes of project (ORDER BY sort_order)
POST	/	classes.js:19	Create class (auto sort_order)
PATCH	/:classId	classes.js:30	Update name/color/hotkey
DELETE	/:classId	classes.js:45	Delete class (CASCADE annotations)

Lưu ý: `hotkey` bị truncate còn 1 ký tự (`normalizeHotkey` function:L7).

3.5 `server/src/routes/images.js`

Thuộc tính	Giá trị	Confidence
---	---	---
Router options	Router({ mergeParams: true })	CONFIRMED
Imports	db, UPLOAD_DIR, multer, sharp, adm-zip, nanoid	CONFIRMED
Callers/Used-by	index.js (mounted /api/projects/:projectId/images)	CONFIRMED
Max file size	100MB / ảnh, 2GB / zip, 20000 ảnh	CONFIRMED

	/ upload	
Định dạng	JPG, PNG, WEBP, BMP	CONFIRMED
Last verified	2026-08-05 (STEP-3.4)	-

Method	Path	File:Line	Mô tả
---	---	---	---
GET	/	images.js:51	List images (kèm class_ids array + thumbnail_url — STEP-1.3)
GET	/:imageId	images.js:64	Get single image + annotations + annotationVersion (STEP-3.4)
POST	/upload	images.js:73	Upload nhiều file ảnh (multer.array)
POST	/upload-zip	images.js:95	Upload ZIP chứa ảnh
PATCH	/:imageId	images.js:143	Update split/status
DELETE	/:imageId	images.js:153	Delete image + xóa file vật lý

[STEP-3.4] GET `/:imageId` response nay trả thêm `annotationVersion: number = COALESCE(MAX(version), 0)` từ `annotation_history` của ảnh. Client dùng giá trị này làm `expectedVersion` khi gọi PUT annotations.

3.6 `server/src/routes/annotations.js`

Thuộc tính	Giá trị	Confidence
---	---	---
Router options	Router({ mergeParams: true })	CONFIRMED
Imports	db, nanoid, pruneAnnotationHistory (từ db.js)	CONFIRMED
Callers/Used-by	index.js (mounted /api/images/:imageId/annotations)	CONFIRMED
Pattern	Replace-all: DELETE + bulk INSERT trong transaction; [STEP-3.2] INSERT snapshot vào annotation_history + prune; [STEP-3.4] Optimistic locking: đọc currentVersion từ annotation_history TRƯỚC transaction, check với expectedVersion từ client body	CONFIRMED
Last verified	2026-08-05 (STEP-3.4)	-

Method	Path	Mô tả	Response
---	---	---	---
PUT	/	Replace toàn bộ annotations; optimistic	200 {annotations: Annotation[],

	lock check; INSERT snapshot annotation_history (version=MAX+1); pruneAnnotationHistory	annotationVersion: number} hoặc 409 {error:'ANNOTATION_CONFLICT', serverVersion:N, message:string}
--	--	--

[STEP-3.4] Optimistic locking:

- Client gửi `expectedVersion` trong body (integer — version client đang cầm từ lần GET gần nhất).
- Server đọc `COALESCE (MAX (version), 0)` từ `annotation_history` TRƯỚC khi vào transaction.
- Nếu `expectedVersion !== currentVersion` → **409 Conflict** (không lưu gì).
- Nếu `expectedVersion` không gửi (undefined/null) → bỏ qua check (backward compat).
- Nếu pass → save bình thường, version tăng lên `currentVersion + 1`.

Version concept: "version của ảnh" = `MAX (version)` trong `annotation_history` (per-save, tăng mỗi lần PUT thành công). **KHÔNG phải** cột `annotations.version` per-row (cột đó chỉ là placeholder từ migration 3.1, chưa được dùng cho locking).

Quan trọng: Không có GET riêng — annotations được trả cùng GET `/images/:imageId` (images.js:64).

Depth-1 callers: index.js (mount), history.js (dùng chung DB nhưng không import annotations.js).

3.7 `server/src/routes/export.js`

Thuộc tính	Giá trị	Confidence
---	---	---
Router options	Router({ mergeParams: true })	CONFIRMED
Imports	db, UPLOAD_DIR, archiver	CONFIRMED
Callers/Used-by	index.js (mounted /api/projects/:projectId/export)	CONFIRMED
Last verified	2026-08-04	-

Method	Path	File:Line	Mô tả			
---	---	---	---			
GET	<code>/split-preview</code>	export.js:85	Preview phân chia train/valid/test trước export			
GET	<code>/</code>	export.js:114	Export ZIP (format=yolo\	coco\ voc, splitMode>manual\ auto)		

Hỗ trợ: YOLO (bbox + OBB quad), COCO JSON, Pascal VOC XML. Stratified split logic (rarest class first).

3.8 `server/src/routes/stats.js`

Method	Path	File:Line	Mô tả
---	---	---	---
GET	<code>/</code>	stats.js:6	Tổng số ảnh, labeled, bySplit, perClass

		annotation count
--	--	------------------

3.9 `server/src/routes/models.js`

Method	Path	File:Line	Mô tả
---	---	---	---
GET	/	models.js:26	List models của project
POST	/upload	models.js:32	Upload file .pt (max 1GB)
DELETE	/:modelId	models.js:43	Delete model + xóa file .pt

3.10 `server/src/routes/autolabel.js`

Thuộc tính	Giá trị	Confidence
---	---	---
Router options	Router({ mergeParams: true })	CONFIRMED
Imports	db, UPLOAD_DIR, MODEL_DIR, child_process.spawn, readline, node:path, nanoid	CONFIRMED
[STEP-1.2] Job storage	DB-backed: <code>createJobInDB()</code> khi POST; <code>syncJobToDB()</code> khi hoàn thành; <code>getJobFromDB()</code> khi GET fall-back. In-memory Map chỉ dùng trong quá trình inference đang chạy.	CONFIRMED
[STEP-1.1] Inference mode mặc định	HTTP fetch → <code>http://127.0.0.1:{INFERENCE_PORT}/predict</code> (FastAPI service thường trực)	CONFIRMED
[STEP-1.1] Rollback mode	<code>USE_LEGACY_INFER=1</code> → dùng spawn infer.py stdin/stdout cũ (giữ nguyên để rollback, KHÔNG có <code>detect_cache</code>)	CONFIRMED
Health gate	Trước mỗi job, gọi <code>GET /health</code> (timeout 3s); trả 503 nếu service chưa sẵn sàng	CONFIRMED
Batch size	32 ảnh/request tới FastAPI; timeout 5 phút/batch	CONFIRMED
[STEP-4.1] <code>detect_cache</code>	<code>CACHE_RAW_CONF=0.01</code> ; <code>getCachedDetections(imageId, modelId)</code> → cache hit: <code>filterRawBoxesByConf(conf)</code> → <code>saveDetections</code> ; cache miss: gọi inference với <code>CACHE_RAW_CONF</code> → <code>saveToCacheDetections()</code> → filter → save	CONFIRMED
Env vars đọc	<code>INFERENCE_PORT</code> (default 8001), <code>USE_LEGACY_INFER</code>	CONFIRMED
Last verified	2026-08-05 (STEP-4.1)	-

Method	Path	File:Line	Mô tả
---	---	---	---
POST	/	autolabel.js:~340	Khởi chạy job inference; tạo row DB ngay; returns <code>{jobId, total}</code> 202
GET	/:jobId	autolabel.js:~400	Poll trạng thái: memory Map → fall-back DB

DELETE	/cache	autolabel.js:~420	Xóa detect_cache entries theo project (+ filter ?imageId=&modelId=). Auth required. Response: {deleted: N}
--------	--------	-------------------	---

Cơ chế mới (STEP-1.1): `checkInferenceHealth()` → `fetch(INFERENCE_URL/predict, batch)` → parse response {classes, results, errors} → `saveDetectionsForImage()` → cập nhật job.

Cơ chế cache (STEP-4.1): Mỗi batch: tách hits (có cache) / misses (cần inference). Hits: load raw_detections từ detect_cache → `filterRawBoxesByConf(conf)` → save. Misses: gọi inference với `CACHE_RAW_CONF=0.01` → `saveToCacheDetections()` → `filterRawBoxesByConf(conf)` → save. Cho phép đổi conf threshold mà không detect lại model.

Cơ chế persist (STEP-1.2): `createJobInDB()` khi POST; HTTP mode `.then()/catch()` → `syncJobToDB()` → `jobs.delete()`. Legacy mode: `setInterval(500ms)` poll job.status → sync khi done/error.

Legacy (USE_LEGACY_INFER=1): `spawn(PYTHON_BIN, [INFER_SCRIPT])` → stdin JSON → readline stdout — giữ nguyên, không xóa. **KHÔNG dùng detect_cache trong legacy mode.**

Depth-1 callers: index.js (mount /api/projects/:pid/auto-label) — không thay đổi mount path.

3.12 `server/src/routes/thumbnails.js` — Thumbnail cache service (STEP-1.3 MỚI)

Thuộc tính	Giá trị	Confidence
---	---	---
Router options	Router() (không mergeParams)	CONFIRMED
Imports	db, UPLOAD_DIR, DATA_DIR, path, fs, sharp	CONFIRMED
Callers/Used-by	index.js (mounted /api/images)	CONFIRMED
Cache dir	DATA_DIR/thumbnails/ — tạo sẵn khi module load	CONFIRMED
Naming cache file	{imageId}_{size}.jpg	CONFIRMED
Size range	min 32, max 600, default 300	CONFIRMED
Fallback	resize lỗi → serve ảnh gốc (không trả 500)	CONFIRMED
Last verified	2026-08-04 (STEP-1.3)	-

Method	Path	Mô tả
---	---	---
GET	/:id/thumb	Lazy thumbnail: cache hit → serve; ảnh gốc nhỏ hơn size → serve gốc; else resize+cache+serve

Query params: ?size=300 (default 300, min 32, max 600)

Cache-Control: public, max-age=86400 (1 ngày)

Exports: THUMB_DIR (dùng để test/cleanup)

3.11 `server/src/routes/jobs.js` — Job history CRUD (STEP-1.2 MỚI)

Thuộc tính	Giá trị	Confidence
---	---	---
Router options	Router()	CONFIRMED
Imports	db	CONFIRMED
Callers/Used-by	index.js (mounted /api/jobs)	CONFIRMED
Last verified	2026-08-04 (STEP-1.2)	-

Method	Path	File:Line	Mô tả
---	---	---	---
GET	/	jobs.js:~22	List jobs by projectId (query param <code>?projectId=</code>) — ORDER BY created_at DESC
GET	/:id	jobs.js:~31	Get single job by id
DELETE	/:id	jobs.js:~37	Delete job record

Response shape (`normalizeJob`): {id, projectId, status, total, done, created, failed, modelId, error, unmatchedClasses[], createdAt, updatedAt}

3.14 `server/src/routes/history.js` — Annotation History (STEP-3.2 MỚI)

Thuộc tính	Giá trị	Confidence
---	---	---
Router options	Router({ mergeParams: true })	CONFIRMED
Imports	db, nanoid, pruneAnnotationHistory (db.js), requireRole (middleware/roles.js)	CONFIRMED
Callers/Used-by	index.js (mounted /api/images/:imageId)	CONFIRMED
Last verified	2026-08-05 (STEP-3.2)	-

Method	Path	Role required	Mô tả
---	---	---	---
GET	/history	any authenticated	List history entries (id, version, actor_id, created_at, actor_name, actor_username) — KHÔNG kèm snapshot
POST	/history/:version/revert	reviewer, admin	Revert về version chỉ định: ghi đè annotations từ snapshot + INSERT history entry mới (audit) + pruneAnnotationHistory

Response GET /history: [{id, version, actor_id, created_at, actor_name, actor_username}] — sắp xếp DESC theo version.

Response POST revert: `{reverted_to_version: N, annotations: Annotation[]}`.

Role guard (AD-A5 Row 7): annotator → 403. Annotator được xem history (GET) nhưng không được revert.

Watch out: Sau revert, 1 history entry mới được tạo (version tăng tiếp) — audit trail liên tục, không bị gián đoạn.

3.15 `server/src/routes/activity.js` — Activity Log (STEP-3.3 MỚI)

Thuộc tính	Giá trị	Confidence
---	---	---
Router options	Router({ mergeParams: true })	CONFIRMED
Imports	db (db.js)	CONFIRMED
Callers/Used-by	index.js (mounted <code>/api/projects/:projectId/activity</code>)	CONFIRMED
Last verified	2026-08-05 (STEP-3.3)	-

Method	Path	Role required	Mô tả
---	---	---	---
GET	/	any authenticated	List activity log của project. LEFT JOIN users để lấy actor_name/actor_color. Phân trang: limit (mặc định 50, max 100) + offset. <code>detail</code> field trả về dưới dạng object (JSON.parse).

Response shape: `[{id, project_id, actor_id, action, detail: object|null, created_at, actor_name, actor_color}]` — ORDER BY created_at DESC.

Actions được ghi (bởi routes khác):

- `image_upload` — images.js (upload single + zip) → `{count, names, source?}`
- `image_delete` — images.js (DELETE) → `{image_id, filename}`
- `split_change` — images.js (PATCH) → `{image_id, from, to}`
- `export` — export.js → `{format, count}`

3.13 `server/src/routes/reviews.js` — Review workflow (STEP-2.3 MỚI)

Thuộc tính	Giá trị	Confidence
---	---	---
Router options	Router({ mergeParams: true })	CONFIRMED
Imports	db, requireRole (middleware/roles.js)	CONFIRMED
Callers/Used-by	index.js (mounted <code>/api/images/:imageId</code>)	CONFIRMED
Last verified	2026-08-04 (STEP-2.3)	-

Method	Path	Role required	Mô tả
---	---	---	---
POST	/submit-review	annotator, admin	draft\ rejected → in_review
POST	/approve	reviewer, admin	in_review → approved
POST	/reject	reviewer, admin	in_review → rejected (body: {comment})

Lifecycle: draft → in_review → approved hoặc rejected → (submit lại) → in_review. Conflict trạng thái sai → 409 REVIEW_STATUS_CONFLICT.

3.16 `server/src/routes/prefill.js` — Prefill bbox tự động (STEP-4.2 MỚI)

Thuộc tính	Giá trị	Confidence
---	---	---
Router options	Router({ mergeParams: true })	CONFIRMED
Imports	db, fetch (node-fetch), requireRole (middleware/roles.js)	CONFIRMED
Callers/Used-by	index.js (mounted /api/projects/:projectId)	CONFIRMED
Last verified	2026-08-05 (STEP-4.2)	-

Method	Path	Role required	Mô tả
---	---	---	---
GET	/images/:imageId/prefill	authenticated	Trả gợi ý bbox; 200+[] nếu đã có annotation; 422 nếu chưa có model mặc định; cache hit/miss inference
PATCH	/default-model	reviewer, admin	Đặt/bỏ model mặc định cho project ({ model_id: string\ null })

Constants: INFERENCE_PORT=8765, CACHE_RAW_CONF=0.01, PREFILL_DEFAULT_CONF=0.4, USE_LEGACY_INFER

Helpers inlined (copy từ autolabel.js — không import):

- getCachedDetections(imageId, modelId) — query bảng detect_cache
- saveToCacheDetections(imageId, modelId, boxes) — insert với raw_boxes JSON, conf=CACHE_RAW_CONF
- filterRawBoxesByConf(boxes, conf) — filter array theo threshold

- `buildSuggestions(projectId, classes, boxes, conf)` — map `class_name` → `class_id` từ DB, trả `SuggestedBox[]`
- `checkInferenceHealth()` — 3s timeout GET
`http://localhost:${INFERENCE_PORT}/health`

Error codes trả về:

- 422 NO_DEFAULT_MODEL — project chưa có `default_model_id`
- 422 DEFAULT_MODEL_NOT_FOUND — `model_id` đã lưu không còn trong DB
- 400 INVALID_MODEL_ID — PATCH body thiếu/sai kiểu
- 503 INFERENCE_NOT_READY — service inference không phản hồi (cache miss, non-legacy mode)
- 503 INFERENCE_UNAVAILABLE_LEGACY — legacy mode, không gọi inference được

4. Module map — Client

4.1 `client/src/App.tsx` — Router root

Route	Component	Mô tả
---	---	---
/	ProjectsPage	Danh sách projects
/projects/:projectId	ProjectDetailPage	Chi tiết project, image grid
/projects/:projectId/annotate/:imageId	AnnotatorPage	Canvas annotator

4.2 `client/src/api.ts` — API client

Thuộc tính	Giá trị	Confidence
---	---	---
Pattern	Thin fetch wrapper (<code>request<T></code>) — không có state	CONFIRMED
Callers/Used-by	ProjectsPage, ProjectDetailPage, AnnotatorPage	CONFIRMED
Auth	JWT Bearer + httpOnly cookie (STEP-2.2)	CONFIRMED
Last verified	2026-08-05 (STEP-4.2)	-

Hàm public:

```
listProjects, createProject, getProject, deleteProject
listClasses, createClass, updateClass, deleteClass
listImages, getImage, uploadImages, uploadZip, updateImage, deleteImage
saveAnnotations(imageId, annotations, expectedVersion?) ← [STEP-3.4] thêm tham số
expectedVersion
→ trả { annotations: Annotation[], annotationVersion: number } (thay vì Annotation[] cũ)
→ throw Error('ANNOTATION_CONFLICT') nếu server trả 409
exportUrl, getSplitPreview
getStats
listModels, uploadModel, deleteModel
```

```

startAutoLabel, getAutoLabelJob
markImageDone, unmarkImageDone
submitReview, approveReview, rejectReview
getPrefill(projectId, imageId, conf?) ← [STEP-4.2] MỚI — trả { suggestions:
SuggestedBox[] }
setDefaultModel(projectId, modelId) ← [STEP-4.2] MỚI — PATCH default-model,
modelId=null để bỏ

```

[STEP-4.2] `getPrefill` silently ignores 422 (no model) ở AnnotatorPage — prefill là tính năng nền, không block UX. `setDefaultModel` dùng trong ProjectDetailPage bởi reviewer/admin.

4.3 `client/src/types.ts` — TypeScript types

Các interface chính:

- **Project** — id, name, description, created_at, image_count, labeled_count, class_count, **default_model_id?: string | null** (STEP-4.2)
- **ClassLabel** — id, project_id, name, color, sort_order, hotkey
- **ImageItem** — id, project_id, filename, original_name, width, height, split, status, created_at, class_ids[], thumbnail_url? (STEP-1.3), review_status?, review_comment?, reviewed_by?, reviewed_at? (STEP-2.3)
- **Annotation** — id, image_id, class_id, x, y, w, h, type, points
- **ImageWithAnnotations** — ImageItem + annotations[] + **annotationVersion: number** (STEP-3.4 — version dùng cho optimistic locking)
- **ModelInfo** — id, project_id, filename, original_name, created_at
- **AutoLabelJob** — status, total, done, created, failed, error, unmatchedClasses[]
- **`SuggestedBox`** (STEP-4.2 MỚI) — class_id, x, y, w, h, type: 'bbox'|'quad', conf, points?

4.4 `client/src/pages/AnnotatorPage.tsx` — Canvas editor

Thuộc tính	Giá trị	Confidence
---	---	---
Callers/Used-by	App.tsx (route /projects/:pid/annotate/:iid)	CONFIRMED
API calls	api.listClasses, api.listImages, api.getImage, api.saveAnnotations, api.submitReview, api.approveReview, api.rejectReview	CONFIRMED
Autosave	debounce 600ms sau mỗi thay đổi box	CONFIRMED
Tools	bbbox (draw rect), quad (4-point OBB)	CONFIRMED
Hotkeys	Arrow ← → navigate, Delete xóa box	CONFIRMED
State: boxes	Box[] (local) — sync lên server khi dirty	CONFIRMED
State: undo	KHÔNG có undo/redo hiện tại	CONFIRMED
[STEP-3.4] annotationVersionRef	<code>useRef<number>(0)</code> — lưu version annotation hiện tại (từ <code>img.annotationVersion</code> khi load). Dùng làm <code>expectedVersion</code> khi gọi <code>saveAnnotations()</code> . Cập nhật thành <code>result.annotationVersion</code> sau mỗi save thành công.	CONFIRMED

[STEP-3.4] 409 handling	<code>scheduleSave</code> catch <code>Error('ANNOTATION_CONFLICT')</code> → alert + reload image + cập nhật <code>annotationVersionRef</code> . Lỗi khác → trạng thái <code>dirty</code> để user retry.	CONFIRMED
[STEP-4.2] Prefill	Khi load ảnh xong: nếu <code>imgBoxes.length === 0</code> → gọi <code>api.getPrefill()</code> background. Trả <code>SuggestedBox[]</code> → map qua <code>suggestionToBox()</code> → <code>setBoxes()</code> (KHÔNG gọi <code>scheduleSave</code>). State <code>prefillLoading</code> + <code>prefillCount</code> hiển thị banner. User action đầu tiên (<code>updateBoxes</code>) reset <code>prefillCount</code> → 0.	CONFIRMED
[STEP-4.2] cancelled flag	<code>useEffect</code> dùng <code>let cancelled = false</code> → cleanup <code>cancelled = true</code> khi unmount/imageId change — tránh <code>setState</code> trên unmounted component.	CONFIRMED
Last verified	2026-08-05 (STEP-4.2)	-

4.5 `client/src/pages/ProjectDetailPage.tsx` — Image grid

API calls	<code>listClasses</code> , <code>listImages</code> , <code>uploadImages</code> , <code>uploadZip</code> , <code>updateImage</code> , <code>deleteImage</code> , <code>**listModels**</code> [STEP-4.2], <code>**setDefaultModel**</code> [STEP-4.2]
---	---
Features	Filter by status/split/class/search, pagination (page×pageSize), drag-drop upload
Modals	<code>StatsPanel</code> , <code>ExportModal</code> , <code>AutoLabelModal</code>
[STEP-4.2] Default model UI	Section "Model mặc định (Prefill)" trong side panel: list models, nút "Đặt mặc định" / badge "★ Mặc định" / nút "Bỏ mặc định". Chỉ reviewer/admin mới thấy (role check).

4.6 Components

Component	Callers	Mô tả
---	---	---
<code>Logo.tsx</code>	<code>App.tsx</code>	KZTEK logo SVG inline
<code>StatsPanel.tsx</code>	<code>ProjectDetailPage</code>	Chart/bảng stats project
<code>ExportModal.tsx</code>	<code>ProjectDetailPage</code>	UI chọn format/split, trigger export
<code>AutoLabelModal.tsx</code>	<code>ProjectDetailPage</code>	Chọn model, confidence, scope; poll job progress

5. Module map — Python Inference

5.1 `server/src/python/infer.py` — Legacy (GIỮ LẠI để rollback)

Thuộc tính	Giá trị	Confidence
---	---	---
Protocol	stdin: JSON payload → stdout: streaming NDJSON (1 line/ảnh) + <code>{"done":true}</code>	CONFIRMED
Caller	<code>autolabel.js runInference()</code> (khi <code>USE_LEGACY_INFER=1</code>)	CONFIRMED
Dependencies	ultralytics (import lazy bên trong <code>main()</code>)	CONFIRMED
Model load	<code>YOLO(model_path)</code> — load từ disk mỗi lần spawn	CONFIRMED
Output bbox	<code>{class_index, type:"bbox", x, y, w, h}</code> (pixel coords)	CONFIRMED
Output quad	<code>{class_index, type:"quad", points:[{x,y}×4]}</code> (OBB)	CONFIRMED
Error handling	Per-image try/except → <code>{image_id, error, classes}</code>	CONFIRMED
Status	GIỮ NGUYÊN — rollback bằng <code>USE_LEGACY_INFER=1</code>	CONFIRMED
Last verified	2026-08-04	-

Input format (stdin):

```
{
  "model_path": "/abs/path/model.pt",
  "conf": 0.25,
  "images": [{"id": "nanoid", "path": "/abs/path/img.jpg"}, ...]
}
```

5.2 `server/src/python/inference\_service.py` — FastAPI thường trực (STEP-1.1 MỚI)

Thuộc tính	Giá trị	Confidence
---	---	---
Framework	FastAPI + uvicorn	CONFIRMED
Port	<code>127.0.0.1:{INFERENCE_PORT}</code> (default 8001)	CONFIRMED
LRU cache	<code>LRUModelCache(max_size=3)</code> — tối đa 3 model giữ trong RAM	CONFIRMED
Caller	<code>index.js startInferenceService()</code> — spawn khi Node start	CONFIRMED
Dependencies	fastapi, uvicorn, ultralytics (lazy import bên trong <code>load_model</code>)	CONFIRMED
Lifecycle	<code>index.js</code> spawn/kill; không auto-	CONFIRMED

	restart khi crash (STEP log Service stopped)	
Env vars	INFERENCE_PORT	CONFIRMED
Last verified	2026-08-04 (STEP-1.1)	-

Endpoints:

Method	Path	Mô tả
---	---	---
GET	/health	Trả {status, loaded_models[], uptime_s}
POST	/warmup	Preload model vào LRU cache. Body: {model_path}
POST	/predict	Batch inference. Body: {model_path, conf, iou, images:[{id,path}]}

Response `/predict`:

```
{
  "classes": ["class_name_0", ...],
  "results": [{ "image_id": "nanoid", "detections":
    [{ "class_index": 0, "type": "bbox", "x": ..., "y": ..., "w": ..., "h": ... } ] },
  "errors": [{ "image_id": "nanoid", "error": "message" } ]
}
```

Output bbox: {class_index, conf, type:"bbox", x, y, w, h} — **STEP-4.1:** thêm conf (float 0-1) để Node.js lưu raw detections vào detect_cache

Output quad (OBB): {class_index, conf, type:"quad", points:[{x,y}×4]} — **STEP-4.1:** thêm conf

Error handling: Per-image try/except → gộp vào errors[] (không crash toàn batch)

Backward compat: saveDetectionsForImage() trong Node.js không đọc conf (ignore) — KHÔNG break existing behavior. conf chỉ được dùng bởi filterRawBoxesByConf() khi lấy từ detect_cache.

6. DB Schema thực tế

Nguồn: server/src/db.js — đọc trực tiếp CREATE TABLE statements. Confidence: CONFIRMED toàn bộ.

Bảng `projects`

Cột	Kiểu	Ràng buộc
---	---	---
id	TEXT	PRIMARY KEY
name	TEXT	NOT NULL
description	TEXT	DEFAULT "
created_at	TEXT	NOT NULL DEFAULT datetime('now')
default_model_id	TEXT	nullable, soft-ref→models(id) — [STEP-4.2] model mặc định cho prefill tự động

[STEP-4.2] `default_model_id` là soft reference (không dùng FK constraint trong SQLite ALTER TABLE). Application layer (`prefill.js`) tự validate model còn tồn tại. Khi model bị xóa: không có CASCADE tự động → `prefill.js` trả 422 `DEFAULT_MODEL_NOT_FOUND`, reviewer/admin cần chọn lại model mặc định.

Bảng `classes`

Cột	Kiểu	Ràng buộc
---	---	---
id	TEXT	PRIMARY KEY
project_id	TEXT	NOT NULL, FK→projects(id) ON DELETE CASCADE
name	TEXT	NOT NULL
color	TEXT	NOT NULL
sort_order	INTEGER	NOT NULL DEFAULT 0
hotkey	TEXT	nullable (migration nếu chưa có)

Index: `idx_classes_project` ON `classes(project_id)`

Bảng `images`

Cột	Kiểu	Ràng buộc	
---	---	---	
id	TEXT	PRIMARY KEY	
project_id	TEXT	NOT NULL, FK→projects(id) ON DELETE CASCADE	
filename	TEXT	NOT NULL (tên file trên disk, nanoid+ext)	
original_name	TEXT	NOT NULL (tên gốc khi upload)	
width	INTEGER	NOT NULL	
height	INTEGER	NOT NULL	
split	TEXT	NOT NULL DEFAULT 'train' (train/valid/test)	
status	TEXT	NOT NULL DEFAULT 'unlabeled' (unlabeled/labeled)	
created_at	TEXT	NOT NULL DEFAULT datetime('now')	
review_status	TEXT	NOT NULL DEFAULT 'draft' — STEP-2.3 MỚI (draft→in_review→approved\rejected)	
review_comment	TEXT	nullable — STEP-2.3 MỚI (lý do reject)	
reviewed_by	INTEGER	FK→users(id), nullable — STEP-2.3 MỚI	
reviewed_at	TEXT	nullable — STEP-2.3 MỚI	
completed_at	TEXT	nullable — STEP-3.5 MỚI (ISO datetime, NULL = chưa done)	
completed_by	INTEGER	FK→users(id), nullable —	

STEP-3.5 MỚI (người đã mark done)

Index: `idx_images_project` ON `images(project_id)`, `idx_images_review_status` ON `images(review_status)` (STEP-2.3), `idx_images_completed_at` ON `images(completed_at)` (STEP-3.5)

ADR (done status v3): dùng 2 cột riêng `completed_at/completed_by`, KHÔNG mở rộng enum `status` — tách biệt "có annotation" (`status=labeled`) và "annotator xác nhận xong" (`completed_at NOT NULL`).

Gate submit-review (STEP-3.5): `submit-review` PHẢI có `completed_at IS NOT NULL` — trả 409 `IMAGE_NOT_COMPLETED` nếu chưa done.

Bảng `annotations`

Cột	Kiểu	Ràng buộc
---	---	---
id	TEXT	PRIMARY KEY
image_id	TEXT	NOT NULL, FK→images(id) ON DELETE CASCADE
class_id	TEXT	NOT NULL, FK→classes(id) ON DELETE CASCADE
x	REAL	NOT NULL (pixel, top-left)
y	REAL	NOT NULL (pixel, top-left)
w	REAL	NOT NULL (pixel width)
h	REAL	NOT NULL (pixel height)
type	TEXT	NOT NULL DEFAULT 'bbox' (bbox/quad) — migration nếu chưa có
points	TEXT	nullable, JSON string <code>[[{x,y}×4]]</code> cho type=quad — migration nếu chưa có
created_at	TEXT	NOT NULL DEFAULT datetime('now')
version	INTEGER	NOT NULL DEFAULT 0 — STEP-3.1 MỚI (optimistic locking, dùng ở STEP-3.4)

Index: `idx_annotations_image` ON `annotations(image_id)`

Bảng `annotation\_history` — STEP-3.1 MỚI

Cột	Kiểu	Ràng buộc
---	---	---
id	INTEGER	PRIMARY KEY AUTOINCREMENT
image_id	TEXT	NOT NULL, FK→images(id) ON DELETE CASCADE
version	INTEGER	NOT NULL
snapshot	TEXT	NOT NULL — JSON toàn bộ annotations của ảnh tại thời điểm save (SNAPSHOT, không diff — ADR AD-5)
actor_id	INTEGER	FK→users(id), nullable
created_at	TEXT	NOT NULL DEFAULT datetime('now')

Index: `idx_ann_history_image, idx_ann_history_img_ver(image_id, version DESC)`. **Retention:** tối đa 200 version/ảnh qua `pruneAnnotationHistory(imageId)` (export từ `db.js`, gọi bởi STEP-3.2).

Bảng `activity\_log` — STEP-3.1 MỚI

Cột	Kiểu	Ràng buộc
---	---	---
id	INTEGER	PRIMARY KEY AUTOINCREMENT
project_id	TEXT	NOT NULL, FK→projects(id) ON DELETE CASCADE
actor_id	INTEGER	FK→users(id), nullable
action	TEXT	NOT NULL (VD: 'upload', 'export', 'split_change')
detail	TEXT	nullable, JSON tùy ý
created_at	TEXT	NOT NULL DEFAULT datetime('now')

Index: `idx_activity_log_project, idx_activity_log_action`. Route ghi log: STEP-3.3 (bảng đã tạo, chưa có route ghi ở STEP-3.1).

⚠ **Lưu ý:** `actor_id` không có `ON DELETE CASCADE/SET NULL` (mặc định NO ACTION) — không thể hard-delete user đã có history/activity_log khi `foreign_keys=ON`. Dùng `is_active=0` (soft-delete, đã có từ STEP-2.2) thay vì xóa thật.

Bảng `models`

Cột	Kiểu	Ràng buộc
---	---	---
id	TEXT	PRIMARY KEY
project_id	TEXT	NOT NULL, FK→projects(id) ON DELETE CASCADE
filename	TEXT	NOT NULL (tên file .pt trên disk)
original_name	TEXT	NOT NULL (tên gốc khi upload)
created_at	TEXT	NOT NULL DEFAULT datetime('now')

Index: `idx_models_project ON models(project_id)`

Bảng `detect\_cache` — STEP-4.1 MỚI

Cột	Kiểu	Ràng buộc
---	---	---
id	INTEGER	PRIMARY KEY AUTOINCREMENT
image_id	TEXT	NOT NULL, FK→images(id) ON DELETE CASCADE
model_id	TEXT	NOT NULL, FK→models(id) ON DELETE CASCADE
raw_detections	TEXT	NOT NULL — JSON: <code>{classes:string[], boxes:[{class_index, conf, x, y, w, h, type, points?}]}</code> — conf gốc, KHÔNG áp threshold
created_at	TEXT	NOT NULL DEFAULT datetime('now')

Index: `idx_detect_cache_img_model` UNIQUE ON `detect_cache(image_id, model_id)` — 1 entry/cặp ảnh-model; `idx_detect_cache_image, idx_detect_cache_model`

Cascade: ON DELETE CASCADE cả `image_id` lẫn `model_id` → cache tự xóa khi ảnh hoặc model bị xóa.

Thiết kế: Cache theo `(image_id, model_id)` — không cache theo `conf/iou`. `raw_detections` lưu ở `CACHE_RAW_CONF=0.01` để cho phép đổi `conf` user mà không detect lại. `IoU=0.45` (default inference service) — NMS đã áp tại inference, không thể thay đổi từ cache.

INSERT OR REPLACE: `saveToCacheDetections()` dùng UPSERT — detect lại cùng ảnh+model sẽ cập nhật cache.

Bảng `jobs` — STEP-1.2 MỚI

Cột	Kiểu	Ràng buộc
---	---	---
id	TEXT	PRIMARY KEY
project_id	TEXT	NOT NULL (no FK — không CASCADE khi xóa project, job history giữ lại)
status	TEXT	NOT NULL DEFAULT 'pending' (pending/running/done/error)
total_images	INTEGER	NOT NULL DEFAULT 0
processed	INTEGER	NOT NULL DEFAULT 0 (ảnh đã xử lý xong)
created_annotations	INTEGER	NOT NULL DEFAULT 0
failed	INTEGER	NOT NULL DEFAULT 0
model_id	TEXT	nullable
error_msg	TEXT	nullable
unmatched_classes	TEXT	NOT NULL DEFAULT '[]' (JSON array tên class không map được)
created_at	TEXT	NOT NULL DEFAULT datetime('now')
updated_at	TEXT	NOT NULL DEFAULT datetime('now')

Indexes: `idx_jobs_project` ON `jobs(project_id)`, `idx_jobs_status` ON `jobs(status)`

Startup behavior: Khi server khởi động, mọi row có `status IN ('running', 'pending')` → cập nhật thành `'error'` với `error_msg='Server restarted while job was in progress'`.

7. API Endpoints

| Tất cả CONFIRMED từ source (index.js route mounting + từng route file).

Projects

Method	Path	File:Line	Response
---	---	---	---
GET	<code>/api/projects</code>	<code>projects.js:9</code>	Project[] (kèm counts)
POST	<code>/api/projects</code>	<code>projects.js:20</code>	Project (201)
GET	<code>/api/projects/:id</code>	<code>projects.js:33</code>	Project
PATCH	<code>/api/projects/:id</code>	<code>projects.js:39</code>	Project
DELETE	<code>/api/projects/:id</code>	<code>projects.js:48</code>	204

Classes

Method	Path	File:Line	Response
---	---	---	---
GET	/api/projects/:pid/classes	classes.js:13	ClassLabel[]
POST	/api/projects/:pid/classes	classes.js:19	ClassLabel (201)
PATCH	/api/projects/:pid/classes/:cid	classes.js:30	ClassLabel
DELETE	/api/projects/:pid/classes/:cid	classes.js:45	204

Images

Method	Path	File:Line	Response
---	---	---	---
GET	/api/projects/:pid/images	images.js:51	ImageItem[]
GET	/api/projects/:pid/images/:iid	images.js:64	ImageWithAnnotations
POST	/api/projects/:pid/images/upload	images.js:73	ImageItem[] (201)
POST	/api/projects/:pid/images/upload-zip	images.js:95	{created, skipped} (201)
PATCH	/api/projects/:pid/images/:iid	images.js:143	ImageItem
POST	/api/projects/:pid/images/:iid/mark-done	images.js:~200	ImageItem (completed_at, completed_by set) — STEP-3.5 MỚI
DELETE	/api/projects/:pid/images/:iid/mark-done	images.js:~220	ImageItem (completed_at=null, completed_by=null) — STEP-3.5 MỚI; annotator chỉ bỏ của mình (403 nếu người khác)
DELETE	/api/projects/:pid/images/:iid	images.js:~253	204

Annotations

Method	Path	File:Line	Response
---	---	---	---
PUT	/api/images/:iid/annotations	annotations.js:14	200 {annotations: Annotation[], annotationVersion: number} — [STEP-3.4] hoặc 409 {error: 'ANNOTATION_CONFLICT', serverVersion: N, message: str} khi conflict. Body: {annotations, expectedVersion?}.

Annotation History (STEP-3.2 MỚI)

Method	Path	File:Line	Role	Response
---	---	---	---	---
GET	/api/images/:iid/history	history.js:2	any	[[id, version,

		6	authenticate d	<code>actor_id, created_at, actor_name, actor_username}]</code> — không có snapshot
POST	<code>/api/images/:iid/history/:version/revert</code>	history.js:4 7	reviewer, admin	<code>{reverted_to_version : N, annotations: Annotation[]}</code>

Export

Method	Path	File:Line	Response
---	---	---	---
GET	<code>/api/projects/:pid/export/split-preview</code>	export.js:85	SplitPreview JSON
GET	<code>/api/projects/:pid/export</code>	export.js:114	ZIP stream (attachment)

Query params export: `format=yolo\|coco\|voc, splitMode>manual\|auto, trainRatio=0.8`

Stats

Method	Path	File:Line	Response
---	---	---	---
GET	<code>/api/projects/:pid/stats</code>	stats.js:6	ProjectStats JSON

Models

Method	Path	File:Line	Response
---	---	---	---
GET	<code>/api/projects/:pid/models</code>	models.js:26	ModelInfo[]
POST	<code>/api/projects/:pid/models/upload</code>	models.js:32	ModelInfo (201)
DELETE	<code>/api/projects/:pid/models/:mid</code>	models.js:43	204

Auto-label

Method	Path	File:Line	Response
---	---	---	---
POST	<code>/api/projects/:pid/auto-label</code>	autolabel.js:~340	{jobId, total} (202)
GET	<code>/api/projects/:pid/auto-label/:jid</code>	autolabel.js:~400	AutoLabelJob (memory → DB fallback)
DELETE	<code>/api/projects/:pid/auto-label/cache</code>	autolabel.js:~420	<code>{deleted: N}</code> — xóa detect_cache entries của project; filter: ?imageId=, ?modelId= (tùy chọn); auth required (mọi role); 404 nếu project không tồn tại — STEP-4.1 MỚI

Jobs (STEP-1.2 MỚI)

Method	Path	File:Line	Response
---	---	---	---
GET	/api/jobs?projectId=:pid	jobs.js:~22	JobRecord[]
GET	/api/jobs/:id	jobs.js:~31	JobRecord
DELETE	/api/jobs/:id	jobs.js:~37	{ok:true}

Thumbnails (STEP-1.3 MỚI)

Method	Path	File:Line	Response
---	---	---	---
GET	/api/images/:id/thumb	thumbnails.js:29	JPEG binary (thumbnail or original) — 200; 404 nếu image/file không tồn tại

Query params: ?size=300 (default 300px, min 32, max 600)

Cache-Control: public, max-age=86400

8. Dependency graph

Server dependencies (package.json — CONFIRMED)

Package	Version	Dùng ở
---	---	---
express	^4.21.0	index.js, tất cả routes
better-sqlite3	^9.6.0	db.js
cors	^2.8.5	index.js
multer	^1.4.5-lts.1	images.js, models.js
sharp	^0.32.6	images.js (metadata width/height, resize)
adm-zip	^0.6.0	images.js (unzip upload)
archiver	^7.0.1	export.js (tạo zip export)
nanoid	^5.0.7	projects, classes, images, annotations, models, autolabel

Client dependencies (package.json — CONFIRMED)

Package	Version	Dùng ở
---	---	---
react	^18.3.1	tất cả components/pages
react-dom	^18.3.1	main.tsx
react-router-dom	^6.26.2	App.tsx, các pages
vite	^5.4.6	dev/build tool
typescript	^5.6.2	type checking

Python dependencies

Package	Dùng ở	Confidence
---	---	---
ultralalytics	infer.py, inference_service.py (YOLO model, predict — lazy import)	CONFIRMED
fastapi	inference_service.py (FastAPI app, endpoints, Pydantic schemas)	CONFIRMED
uvicorn	inference_service.py (ASGI server, entrypoint)	CONFIRMED

9. Điểm sẽ bị ảnh hưởng bởi plan cải tiến

Mục này là "blast radius map" cho 23 bước trong PLAN-MASTER.md.
Confidence: CONFIRMED từ phân tích source + plan.

Phase 1.1 — Inference service FastAPI (thường trực) — ✔ HOÀN THÀNH

Files đã thay đổi (STEP-1.1):

- `server/src/routes/autolabel.js` — Refactor: thêm `runInferenceHTTP()` gọi FastAPI, giữ `runInference()` cũ cho rollback (USE_LEGACY_INFER=1). Health gate trước mỗi job → 503 khi service down.
- `server/src/index.js` — Thêm `startInferenceService()` / `stopInferenceService()` lifecycle.
- `server/src/python/inference_service.py` — FastAPI app mới, LRU cache 3 model.
- `server/src/python/infer.py` — GIỮ NGUYÊN (rollback).
- `server/README.md` — Thêm mục "Chạy inference service", env vars, cấu trúc Python inference.

Depth-1 callers: `index.js` (mount `autolabelRouter`) — không đổi mount path.

Watch out: `client/api.ts` `startAutoLabel` + `getAutoLabelJob` không đổi (same REST interface).

Phase 1.2 — Persist bảng `jobs` — ✔ HOÀN THÀNH

Files đã thay đổi (STEP-1.2):

- `server/src/db.js` — thêm `CREATE TABLE IF NOT EXISTS jobs (...)` + 2 indexes + startup cleanup (UPDATE running/pending → error)
- `server/src/routes/autolabel.js` — thêm `createJobInDB`, `syncJobToDB`, `getJobFromDB`; POST handler tạo DB row + sync khi hoàn thành; GET handler fall-back DB
- `server/src/routes/jobs.js` — **MỚI:** GET/GET/:id/DELETE /api/jobs
- `server/src/index.js` — import + mount `jobsRouter` tại `/api/jobs`
- `server/migrations/001_create_jobs.sql` — **MỚI:** migration SQL (tham khảo)

Depth-1 callers: `client/src/api.ts` (`getAutoLabelJob`) — REST interface giữ nguyên, không cần sửa client.

Watch out: `jobs` table không có FK tới `projects` — khi xóa project, jobs không bị cascade. Thiết kế có chủ ý (giữ job history). Phase sau nếu cần cleanup → `DELETE FROM jobs WHERE project_id=?` trong `routes/projects.js` khi `DELETE` project.

Phase 1.3 — Thumbnail service — ✔ HOÀN THÀNH

Files đã thay đổi (STEP-1.3):

- `server/src/routes/thumbnails.js` — **MỚI:** lazy thumbnail endpoint `GET /api/images/:id/thumb`, cache `DATA_DIR/thumbnails/{id}_{size}.jpg`
- `server/src/index.js` — import + mount `thumbnailsRouter` tại `/api/images`
- `server/src/routes/images.js` — `GET /` thêm field `thumbnail_url` vào response
- `client/src/types.ts` — thêm `thumbnail_url?: string` vào `ImageItem`
- `client/src/pages/ProjectDetailPage.tsx` — image grid dùng `img.thumbnail_url` thay `/uploads/...`

Depth-1 callers:

- `client/src/pages/ProjectDetailPage.tsx` → dùng `img.thumbnail_url` (thêm)
- `client/src/pages/AnnotatorPage.tsx` → không thay đổi (dùng `getImage` riêng, không qua list)

Watch out: `THUMB_DIR` tạo khi module load — nếu `DATA_DIR` không ghi được sẽ crash lúc server start. Cần đảm bảo `server/data/` có write permission.

Phase 2.2 — Auth implementation — ✔ HOÀN THÀNH

Files bị ảnh hưởng:

- `server/src/db.js` — bảng `users` (INTEGER AUTOINCREMENT id, username, password_hash, display_name, role, color, created_at, is_active)
- `server/src/index.js` — mount `auth.js/users.js`, middleware `auth`
- Tạo mới: `server/src/routes/auth.js`, `server/src/routes/users.js`, `server/src/middleware/`, `server/src/lib/`
- `client/src/api.ts`, `client/src/types.ts`, `client/src/App.tsx`, `client/src/pages/LoginPage.tsx`
- Test: `tests/auth.test.js` — 87 passed theo ma trận AD-A5 (2 điều kiện CTO #A1/#A2 đạt)

Phase 2.3 — Review workflow — ✔ HOÀN THÀNH

Files bị ảnh hưởng:

- `server/src/db.js` — `m004_add_review_status()`: 4 cột mới trên `images` (`review_status`, `review_comment`, `reviewed_by`, `reviewed_at`) + index
- Tạo mới: `server/src/routes/reviews.js` (xem §3.13)
- `server/src/index.js` — mount `reviewsRouter` tại `/api/images/:imageId`
- Test: `tests/auth.test.js` Row 8 — 8 test case pass (submit-review, approve, reject, comment, role guard, 401)

Đã xử lý ở STEP-3.5: `submit-review` trong `reviews.js` đã check `completed_at IS NOT NULL` — trả 409 `IMAGE_NOT_COMPLETED` nếu chưa done.

Phase 3.1 — DB Migration (ĐỤNG SCHEMA HIỆN CÓ) — ✔ HOÀN THÀNH

Files bị ảnh hưởng:

- `server/src/db.js` — `m005_phase3_schema()`: cột `annotations.version` DEFAULT 0, bảng `annotation_history` (snapshot), `activity_log`; auto-backup `.bak-{ts}` (giữ 5 bản) + verify row count trước/sau (throw nếu mất dữ liệu — CTO condition #1); export `pruneAnnotationHistory()`.
- Chưa đụng `server/src/routes/annotations.js` — route UPDATE version + INSERT history là việc của STEP-3.2, bước này CHỈ tạo schema.

Kết quả verify row count (thực tế đã chạy): projects 2→2, classes 4→4, images 1→1, annotations 0→0, models 0→0, jobs 0→0, users 1→1 — không mất dữ liệu ở bảng nào.

Depth-1 callers sẽ break nếu STEP-3.2 sai: tất cả code đọc/ghi bảng `annotations` (`annotations.js`, `images.js:64`, `export.js`, `stats.js`, `autolabel.js`) — CHƯA đổi ở bước này, cần re-verify khi STEP-3.2 sửa `annotations.js`.

Watch out: `annotation_history.snapshot` là SNAPSHOT JSON (không diff) — STEP-3.2 revert = ghi đè thẳng, không merge. `actor_id` không có ON DELETE CASCADE — không hard-delete user có history.

Phase 3.2 — Annotation history + revert — ✔ HOÀN THÀNH

Files đã thay đổi (STEP-3.2):

- `server/src/routes/annotations.js` — import `pruneAnnotationHistory`; PUT handler: trong transaction thêm `COALESCE(MAX(version), 0) + 1` → INSERT snapshot → gọi `pruneAnnotationHistory(imageId)` ngoài transaction.
- `server/src/routes/history.js` — **MỚI:** GET `/history` (list, no snapshot) + POST `/history/:version/revert` (reviewer/admin, AD-A5 Row 7).
- `server/src/index.js` — import + mount `historyRouter` tại `/api/images/:imageId`.
- `tests/auth.test.js` — Row 7 (3 skip → 6 test thật): 93 passed, 0 failed, 0 skipped.

Depth-1 callers của annotations.js: không thay đổi interface PUT (input/output giống trước), chỉ thêm side-effect ghi history.

Depth-1 của history.js: `index.js` (mount) — client chưa gọi (frontend history panel là việc của phase UI sau).

Watch out cho STEP-3.4: Cần sửa `annotations.js` PUT handler — thêm `req.body.version` conflict check. STEP-3.3 KHÔNG sửa `annotations.js` (chỉ sửa `images.js/export.js`). Không xung đột.

Phase 3.3 — Activity log cấp project — ✔ HOÀN THÀNH

Files đã thay đổi (STEP-3.3):

- `server/src/db.js` — thêm export `logActivity(projectId, actorId, action, detail)`: INSERT non-blocking vào `activity_log`.
- `server/src/routes/activity.js` — **MỚI:** GET `/api/projects/:projectId/activity` (phân trang, LEFT JOIN users, detail dưới dạng object).
- `server/src/routes/images.js` — gọi `logActivity` sau: upload (single + zip) → `image_upload`, PATCH khi split đổi → `split_change`, DELETE → `image_delete`.

- `server/src/routes/export.js` — gọi `logActivity` trước `archive.finalize()` → `export`.
- `server/src/index.js` — mount `activityRouter` tại `/api/projects/:projectId/activity`.
- `tests/auth.test.js` — Row 17: verify `image_upload/split_change/image_delete` log + all roles 200 + unauth 401.

Kết quả test: 101 passed, 0 failed, 0 skipped. Commit: a0f01dc.

Phase 3.4 — Optimistic locking annotation — ✔ HOÀN THÀNH

Files đã thay đổi (STEP-3.4):

- `server/src/routes/annotations.js` — PUT handler: đọc `currentVersion = COALESCE(MAX(version), 0)` từ `annotation_history` TRƯỚC transaction; if `expectedVersion` mismatch → 409; response đổi thành `{annotations, annotationVersion}`.
- `server/src/routes/images.js` — GET `/:imageId` thêm `annotationVersion` vào response (query `annotation_history`).
- `client/src/types.ts` — `ImageWithAnnotations` thêm `annotationVersion: number`.
- `client/src/api.ts` — `saveAnnotations(imageId, annotations, expectedVersion?)` → `{annotations, annotationVersion}`.
- `client/src/pages/AnnotatorPage.tsx` — `annotationVersionRef` ref tracking; `scheduleSave` gửi version, catch 409, reload image.
- `tests/auth.test.js` — Row 18: 12 test case — conflict detection, backward compat, response shape. 113 passed, 0 failed.

Depth-1 callers của PUT annotations: `client/src/api.ts` → `AnnotatorPage.tsx` (`saveAnnotations`).
Không caller server-side nào bị break (`history.js` không gọi PUT annotations route).

Watch out cho Phase 4/5: `annotations.js` PUT response đã thay đổi thành object — bất kỳ code nào gọi PUT và expect `Annotation[]` trực tiếp sẽ break. Hiện chỉ `AnnotatorPage` gọi và đã được cập nhật.

Phase 3.5 — Image done status — ✔ HOÀN THÀNH

Files đã thay đổi (STEP-3.5):

- `server/src/db.js` — `m006_add_image_done_columns()`: ALTER TABLE `images` ADD `completed_at TEXT + completed_by INTEGER REFERENCES users(id) + index`; verify row count trước/sau (CTO condition #1 pattern).
- `server/src/routes/images.js` — 2 endpoint MỚI: POST `/:imageId/mark-done` (tất cả role, set `completed_at=now()/completed_by=req.user.id + logActivity`); DELETE `/:imageId/mark-done` (annotator chỉ bỏ của mình — 403 nếu người khác, reviewer/admin any — set NULL).
- `server/src/routes/reviews.js` — submit-review thêm gate `completed_at IS NOT NULL` → 409 `IMAGE_NOT_COMPLETED` nếu chưa done.
- `client/src/types.ts` — `ImageItem` thêm `completed_at?: string | null, completed_by?: number | null`.
- `client/src/api.ts` — thêm `markImageDone(projectId, imageId)` + `unmarkImageDone(projectId, imageId)`.

- `client/src/pages/AnnotatorPage.tsx` — state `doneBusy`, handlers `handleMarkDone/handleUnmarkDone`, phím tắt D (toggle done), nút "❑ Xong (D)" / "✓ Đã xong" trong toolbar.
- `client/src/pages/ProjectDetailPage.tsx` — type `DoneFilter`, filter dropdown "Đã hoàn thành / Chưa hoàn thành", badge "✓ Xong" màu xanh đậm trên image tile.
- `tests/auth.test.js` — Row 8 (submit-review): thêm mark-done trước mỗi submit; Row 19 (15 test case): gate 409, mark-done, unmark, role guard. 128 passed, 0 failed.

Depth-1 callers:

- `client/api.ts` → `AnnotatorPage.tsx` (`markImageDone/unmarkImageDone`)
- `reviews.js` submit-review gate → caller: `client/api.ts` → `AnnotatorPage.tsx` (`submitReview`)

Watch out cho Phase 4: Phase 4 (detect cache) không đụng bảng

`images/annotations/annotation_history` — không cần cập nhật done logic. Phase 5 (UX) cũng không đụng `completed_at/completed_by`.

Phase 4.1 — Detect cache — ✓ HOÀN THÀNH

Files đã thay đổi (STEP-4.1):

- `server/src/db.js` — `m007_detect_cache()`: CREATE TABLE `detect_cache` (id, image_id FK CASCADE, model_id FK CASCADE, raw_detections TEXT, created_at) + UNIQUE idx (image_id, model_id) + 2 indexes; verify row count trước/sau (CTO condition #1 pattern).
- `server/src/routes/autolabel.js` — thêm `CACHE_RAW_CONF=0.01`; helpers `getCachedDetections()`, `saveToCacheDetections()`, `filterRawBoxesByConf()`; refactor `runInferenceHTTP()`: nhận thêm param `modelId`, tách hit/miss per batch, gọi inference với `CACHE_RAW_CONF` cho miss, lưu raw → filter → save; thêm `DELETE /cache` endpoint; cập nhật call site truyền `model_id`.
- `server/src/python/inference_service.py` — `_predict_single()`: thêm `conf` field vào mỗi box (bbox: `r.bboxes.conf[i]`, OBB: `obb.conf[i]`) — backward compat (Node.js `saveDetectionsForImage` ignore field này).
- `tests/auth.test.js` — Row 20 (4 test case): `DELETE /cache` unauth→401, admin→200, annotator→200, bad project→404. Header comment cập nhật.

Verify row count (m007): CREATE TABLE mới — không có risk mất dữ liệu existing. Pattern giữ nhất quán với m005/m006.

Test kết quả: 133 passed, 0 failed, 0 skipped (128 cũ + 5 mới Row 20).

Depth-1 callers:

- `index.js` (mount `/api/projects/:pid/auto-label`) — không đổi mount path
- `client/api.ts` `startAutoLabel` — REST interface giữ nguyên (không cần sửa client)

Watch out cho STEP-4.2 (Prefill bbox):

- Dùng lại `getCachedDetections(imageId, modelId) + filterRawBoxesByConf(boxes, conf)` từ `autolabel.js` khi mở ảnh chưa label — 2 hàm này có thể import hoặc copy
- Cache key là (image_id, model_id) — STEP-4.2 cần biết `model_id` đang active của project

- `CACHE_RAW_CONF=0.01` đã được dùng để build cache; STEP-4.2 chỉ cần đọc cache + filter, không detect lại
- Nếu cache miss khi mở ảnh → STEP-4.2 tự quyết định: gọi inference ngầm hoặc hiện thông báo "chưa có cache"

Phase 5.1 — Undo/Redo

Files bị ảnh hưởng:

- `client/src/pages/AnnotatorPage.tsx` — thêm `undoStack/redoStack` state, `Ctrl+Z/Ctrl+Y` handler

Phase 6.1 — Model versioning

Files bị ảnh hưởng:

- `server/src/db.js` — `ALTER TABLE models ADD COLUMN metadata JSON`
- `server/src/routes/models.js` — thêm `PATCH` endpoint, `GET` endpoint với `metadata`
- `client/src/types.ts` — mở rộng `ModelInfo`

10. Lịch sử cập nhật

Ngày	Người cập nhật	Nội dung	Commit
---	---	---	---
2026-08-04	senior-developer (STEP-0.1)	Tạo mới — full audit 5 bảng, 8 routes, API endpoints, blast radius map	(STEP-0.1)
2026-08-04	senior-developer (STEP-1.1)	Cập nhật §3.10 autolabel.js (HTTP mode + legacy rollback), thêm §5.2 <code>inference_service.py</code> , cập nhật §8 Python deps, §9 Phase 1.1 DONE	(STEP-1.1)
2026-08-04	senior-developer (STEP-1.2)	Thêm §3.11 jobs.js (MỚI), cập nhật §2 (migrations/ + jobs.js), §3.1 (route <code>/api/jobs</code>), §3.2 (startup cleanup), §3.10 (DB-backed job storage), §6 (jobs table schema), §7 (<code>/api/jobs</code> endpoints), §9 Phase 1.2 DONE	(STEP-1.2)
2026-08-04	junior-developer (STEP-1.3)	Thêm §3.12 thumbnails.js (MỚI), cập nhật §1 (<code>data/thumbnails/</code>), §2 (thư mục + route), §3.1 (mount <code>/api/images</code>), §3.5 (<code>thumbnail_url</code> field), §4.3 (<code>ImageItem.thumbnail_url</code>), §7 (<code>/api/images/:id/thumb</code>), §9 Phase 1.3 DONE	(STEP-1.3)
2026-08-04	tech-lead+cto+em (STEP-2.1)	Auth ADR riêng (<code>ADR-auth-labeling-studio.md</code>), security-audit-stride, CTO+EM APPROVED kèm điều kiện #A1/#A2	(STEP-2.1)
2026-08-04	senior-developer (STEP-2.2)	Bảng <code>users</code> (INTEGER id, role, color), routes <code>auth.js/users.js</code> , middleware, <code>LoginPage</code> — 87 test pass ma trận AD-A5, npm audit sạch (điều kiện #A1/#A2 đạt), §9	(STEP-2.2)

		Phase 2.2 DONE	
2026-08-04	senior-developer (STEP-2.3)	Thêm §3.13 reviews.js (MỚI), cập nhật §6 bảng <code>images</code> (4 cột review), §9 Phase 2.3 DONE — review workflow submit/approve/reject, Row 8 test pass	(STEP-2.3)
2026-08-05	senior-developer (STEP-3.1)	Cập nhật §6 — cột <code>annotations.version</code> , bảng mới <code>annotation_history</code> (snapshot), <code>activity_log</code> ; §9 Phase 3.1 DONE — verify row count trước/sau (0 mất dữ liệu, CTO condition #1 đạt)	(STEP-3.1)
2026-08-05	senior-developer (STEP-3.2)	Thêm §3.14 history.js (MỚI), cập nhật §2 (routes/history.js), §3.1 (mount /api/images/:id history), §3.6 (annotations.js + pruneAnnotationHistory), §7 (Annotation History endpoints), §9 Phase 3.2 DONE — 93 test pass (0 fail, 0 skip)	a4c3519
2026-08-05	junior-developer (STEP-3.3)	Thêm §3.15 activity.js (MỚI), cập nhật §2 (routes/activity.js), §3.1 (mount /api/projects/:id/activity + activityRouter), §3.2 (db.js exports: logActivity), §3.9 (images.js gọi logActivity), §3.10 (export.js gọi logActivity), §9 Phase 3.3 DONE — 101 test pass (0 fail, 0 skip)	a0f01dc
2026-08-05	senior-developer (STEP-3.4)	Cập nhật §3.5 (GET /:imageId trả thêm annotationVersion), §3.6 (annotations.js — optimistic locking: check expectedVersion trước transaction, PUT response đổi từ Annotation[] thành {annotations, annotationVersion}), §4.2 (api.ts — saveAnnotations thêm tham số expectedVersion), §4.3 (ImageWithAnnotations.annotationVersion), §4.4 (AnnotatorPage — annotationVersionRef, 409 handling), §7 (PUT annotations response 409), §9 Phase 3.4 DONE — 113 test pass (0 fail, 0 skip), tsc 0 lỗi	(STEP-3.4)
2026-08-05	senior-developer (STEP-3.5)	Cập nhật §6 (images — 2 cột mới completed_at/completed_by + index), §7 (Images — 2 endpoint MỚI POST/DELETE mark-done; reviews.js gate IMAGE_NOT_COMPLETED), §9 Phase 3.5 DONE — 128 test pass (0 fail, 0 skip), tsc 0 lỗi. Phase 3 HOÀN TOÀN HOÀN THÀNH.	(STEP-3.5)
2026-08-05	senior-developer (STEP-4.1)	Thêm bảng <code>detect_cache</code> (§6), route <code>DELETE /cache</code> (§3.10/§7), cache raw detections theo (image_id, model_id) trong <code>autolabel.js</code> , <code>inference_service.py</code> trả thêm <code>conf</code> , §9 Phase 4.1 DONE — 133 test pass (0 fail, 0 skip)	ac61d98→(STEP-4.1)